# LOCALIZING AWAY FROM THE ZERODIVISORS OF A FIBRE MODULE: THE SPECTRUM IS SEMI-LOCAL ON THE ASSOCIATED PRIMES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

*Editorial note.* Every referenced result of this card lives in an earlier chapter, so each locator points at the verbatim line of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carries the canonical tag.

*Allowed standard prerequisites (editorial).*

- Associated primes of a module over a quotient ring. Algebra, Lemma labelled algebra-lemma-ass-quotient-ring (canonical Stacks tag 05BY); invoked at proof line 1803. Supplies the identification Ass_S(Nbar) = Ass_Sbar(Nbar) for the module Nbar = N/m_R N over Sbar = S/m_R S; the rest of the argument is retained in the counted span.
- Finiteness of the set of associated primes of a finite module over a Noetherian ring. Algebra, Lemma labelled algebra-lemma-finite-ass (canonical Stacks tag 00LC); invoked at proof lines 1804-1806. Supplies only the finiteness of Ass_S(Nbar), which is what makes the enumeration q_1, ..., q_r available.
- Zerodivisors are the union of the associated primes. Algebra, Lemma labelled algebra-lemma-ass-zero-divisors (canonical Stacks tag 00LD); invoked at proof lines 1807-1808. Supplies the identity Sigma = S minus the union of the q_i, which is the translation the counted proof needs in order to describe the localisation.
- Primes of a localisation and primes contained in a given prime. Algebra, Lemma algebra-lemma-spec-localization (tag 00E3) and algebra-lemma-silly (tag 00DS); invoked at proof lines 1809-1813. Supply the description of Spec(Sigma^{-1}S) and the behaviour of primes under localisation; the resulting correspondence with the q_i and the maximality argument are inside the counted span.

*Source reference.* Original lemma 05BY: sources/stacks-chow/tags.txt, source lines 6901–6901. Cross-chapter standard prerequisite invoked once at proof line 1803 to identify associated primes over S with associated primes over S/m_R S; it is a citation of a general algebra fact and replaces nothing in the retained proof.

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*Date*: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 00LC: sources/stacks-chow/tags.txt, source lines 782–782. Cross-chapter standard prerequisite invoked once at proof line 1805 for finiteness of the set of associated primes of a finite module over a Noetherian ring.

*Source reference.* Original lemma 00LD: sources/stacks-chow/tags.txt, source lines 783–783. Cross-chapter standard prerequisite invoked once at proof line 1808 for the description of the zerodivisors of Nbar as the union of its associated primes.

*Source reference.* Original lemma 00E3: sources/stacks-chow/tags.txt, source lines 520–520. Cross-chapter standard prerequisite invoked once at proof lines 1809-1810 for the description of the primes of a localisation; the translation into the primes contained in one of the q_i is the counted step.

*Source reference.* Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Cross-chapter standard prerequisite invoked once at proof lines 1811-1812 for comparing primes under localisation; the maximality conclusion for the maximal ideals of Sigma^{-1}S is retained in the counted span.

**Lemma 0.1.** *Let $R \to S$ be a ring map. Let $N$ be a $S$-module. Assume*

(1) *$R$ is a local ring with maximal ideal $\mathfrak{m}$,*
(2) *$\overline{S} = S/\mathfrak{m}S$ is Noetherian, and*
(3) *$\overline{N} = N/\mathfrak{m}_R N$ is a finite $\overline{S}$-module.*

*Let $\Sigma \subset S$ be the multiplicative subset of elements which are not a zerodivisor on $\overline{N}$. Then $\Sigma^{-1}S$ is a semi-local ring whose spectrum consists of primes $\mathfrak{q} \subset S$ contained in an element of $Ass_S(\overline{N})$. Moreover, any maximal ideal of $\Sigma^{-1}S$ corresponds to an associated prime of $\overline{N}$ over $\overline{S}$.*

*Proof.* Note that $\mathrm{Ass}_S(\overline{N}) = \mathrm{Ass}_{\overline{S}}(\overline{N})$, see Algebra, Lemma 05BY. This is a finite set by Algebra, Lemma 00LC. Say $\{\mathfrak{q}_1, \ldots, \mathfrak{q}_r\} = \mathrm{Ass}_S(\overline{N})$. We have $\Sigma = S \setminus (\bigcup \mathfrak{q}_i)$ by Algebra, Lemma 00LD. By the description of $\mathrm{Spec}(\Sigma^{-1}S)$ in Algebra, Lemma 00E3 and by Algebra, Lemma 00DS we see that the primes of $\Sigma^{-1}S$ correspond to the primes of $S$ contained in one of the $\mathfrak{q}_i$. Hence the maximal ideals of $\Sigma^{-1}S$ correspond one-to-one with the maximal (w.r.t. inclusion) elements of the set $\{\mathfrak{q}_1, \ldots, \mathfrak{q}_r\}$. This proves the lemma. □